

3.4.1 Data Types in Python

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 08 of 24

★ Golden

3.4.1 Data Types in Python ★ Golden

LEARN IT

A **data type** specifies the **kind of data** a variable holds — a whole number, a decimal, text, True/False, or a collection of items. The type decides what you can do with the data (you can divide numbers, but not names). Python has several built-in data types:

Name	Type	Description	Example
Integer	int	Whole numbers without decimals (positive, negative or zero).	age = 16
Floating point	float	Numbers with a decimal point.	height = 5.7
String	str	Text data, enclosed in quotes (" " or ' ').	name = "Sammy" or year = "2026"
Boolean	bool	Logical value: True or False (capital T / F).	is_student = True
List	list	Ordered, changeable collection in square brackets [] .	marks = [85, 90, 75]
Tuple	tuple	Ordered, unchangeable sequence in parentheses () .	coords = (10, 20)
Dictionary	dict	Key : value pairs in curly braces { } .	student = {"name": "Ahmed", "age": 16}
Set	set	Unordered collection of unique items (no duplicates).	vowels = {'a', 'e', 'i'}

Note: the slide abbreviates tuple as "tup"; the real Python type name is `tuple`, as the program below shows. The slides describe dictionaries as unordered; since Python 3.7 a dictionary remembers the order in which keys were added, but you always look items up **by key**, not by position.

WORKED EXAMPLE

Checking every type with type() :

```
age = 16
height = 5.7
name = "Sammy"
year = "2026"
is_student = True
marks = [85, 90, 75]
coords = (10, 20)
student = {"name": "Ahmed", "age": 16}
vowels = {'a', 'e', 'i'}

print(type(age))
print(type(height))
print(type(name), type(year))
print(type(is_student))
print(type(marks))
print(type(coords))
print(type(student))
print(type(vowels))
```

OUTPUT

<class 'int'>
<class 'float'>
<class 'str'> <class 'str'>
<class 'bool'>
<class 'list'>
<class 'tuple'>
<class 'dict'>
<class 'set'>

Notice that year = "2026" is a **string**, not a number, because it is inside quotes.

Working with the collection types:

```
marks = [85, 90, 75]
marks[0] = 88 # lists are changeable
marks.append(60)
print(marks)

coords = (10, 20)
print(coords[0]) # read item at position 0

student = {"name": "Ahmed", "age": 16}
print(student["name"]) # look up by key

nums = {1, 2, 2, 3, 3, 3} # duplicates are removed
print(nums)
print(len(nums))
```

OUTPUT

[88, 90, 75, 60]
10
Ahmed
{1, 2, 3}
3

Positions (called **indexes**) start at **0**, so coords[0] is the first item. Trying to change a tuple causes an error:

```
coords = (10, 20)
coords[0] = 50
```

OUTPUT

Traceback (most recent call last):
 File "tuple_test.py", line 2, in <module>
 coords[0] = 50
 ~~~~~^~
TypeError: 'tuple' object does not support item assignment

REMEMBER IT

Brackets decide the collection: [ ] List (changeable), ( ) Tuple (locked), { } with key:value = Dictionary, { } with single items = Set (unique). Think: a **list** is a shopping list you can edit; a **tuple** is your date of birth — it never changes.

COMMON MISTAKE

True and False must start with a capital letter. true is not a Boolean; Python treats it as an unknown variable name and gives a NameError . Also, "16" (string) and 16 (int) are different: "16" + "1" gives "161", while 16 + 1 gives 17 .

اردو میں سمجھیں

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔ int مکمل اعداد، float اعشاریہ والے اعداد، str متن، اور bool صرف True یا False کے لیے ہے۔ لسٹ (List) تبدیل ہو سکتی ہے، ٹپل (Tuple) تبدیل نہیں ہو سکتا، ڈکشنری (Dictionary) میں کی اور ولیو کے جوڑے ہوتے ہیں، اور سیٹ (Set) میں کوئی چیز دہرائی نہیں جاتی۔

EXAM TIP

Expect: "Name any four data types with examples" or "Differentiate list and tuple". For a difference question, use a two-column table: list — [ ] , changeable, e.g. marks = [85, 90] ; tuple — ( ) , unchangeable, e.g. coords = (10, 20) .

CHECK YOURSELF

1. What is the data type of each? 3.0 , "3" , 3 , False , [3]  
2. Predict the output: print(len({5, 5, 6, 6, 6}))  
3. Which type should store a student's CNIC/B-Form number such as "42101-1234567-1" ? Why?  
Answers: 1. float, str, int, bool, list. 2. 2 (the set keeps only 5 and 6). 3. str , because it contains hyphens and you never do arithmetic on it.

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Next lecture: 3.4.2 Input and Output Handling